

Analyzing the bifurcation of the ENSO replenishment oscillator with delayed feedback

Group member: Eddie Yuan, Leo Luo

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Abstract

This paper investigates the nonlinear dynamics of an El Niño–Southern Oscillation (ENSO) recharge oscillator model incorporating time-delayed feedback. Through a mathematical transformation, the coupled ocean-atmosphere system is reformulated as a time-delayed Van der Pol-Duffing equation, allowing for a rigorous analysis of the system’s dynamical properties. Using the averaging method, we derive analytical conditions for the stability of the zero solution and the onset of Hopf bifurcation, revealing a critical dependence on both the time-delay intensity (κ) and delay time (τ). Specifically, the time-delayed negative feedback (representing Rossby wave propagation and Kelvin wave reflection) can destabilize the equilibrium state, driving the system into quasi-periodic oscillations even when the instantaneous Bjerknes feedback is damping ($\mu > 0$). Numerical simulations confirm that the oscillation amplitude and behaviors are modulated by the delay parameters, with the system exhibiting stable limit cycles or bifurcations under different (κ, τ) regimes. These results demonstrate that the time-delay mechanism is essential for sustaining ENSO self-excited oscillations, providing new insights into the nonlinear interactions between oceanic wave dynamics and atmospheric feedback. The study highlights the importance of delay effects in conceptual ENSO models and their implications for climate predictability.

1 Introduction

The **ENSO** is the dominant recurring inter-annual climate variability pattern in the tropical Pacific region, exerting profound influences on global weather patterns, agriculture, and ecosystems. During El Niño events, abnormal increases in sea surface temperatures (SST) in the central and eastern equatorial Pacific disrupt the Walker circulation and alter global precipitation patterns and storm trajectories. Conversely, La Niña events are characterized by an abnormal decrease in SST in the central and eastern equatorial Pacific, leading to roughly opposite climate anomalies. Despite decades of observational data and ocean-atmosphere coupled numerical models, the core mechanisms underlying ENSO quasi-periodic oscillations remain an active area of research.

The **net damping parameter** (μ) in the ENSO Van der Pol-Duffing Equation derived in the future pages (5)

$$\frac{d^2T}{dt^2} + \omega^2 T + \varepsilon \left[(\mu + 3T^2) \frac{dT}{dt} + R_h T^3 \right] = \varepsilon \Delta$$

is the mutual enhancement (negative damping $\mu < 0$) or suppression (positive damping $\mu > 0$) of SST anomalies and wind field anomalies. In traditional ENSO analysis, an El Niño or La Niña event typically corresponds to negative damping ($\mu < 0$). In the El Niño case, the SST on the East Pacific Ocean near the equator is above normal, causing higher eastern evaporation to decrease the equatorial east wind by decreasing the east-west temperature gradient. This reduces the westward transport of warm equatorial surface water, leading to the buildup of warm water in the eastern Pacific, therefore further increasing the East Pacific SST and leading to an El Niño event.

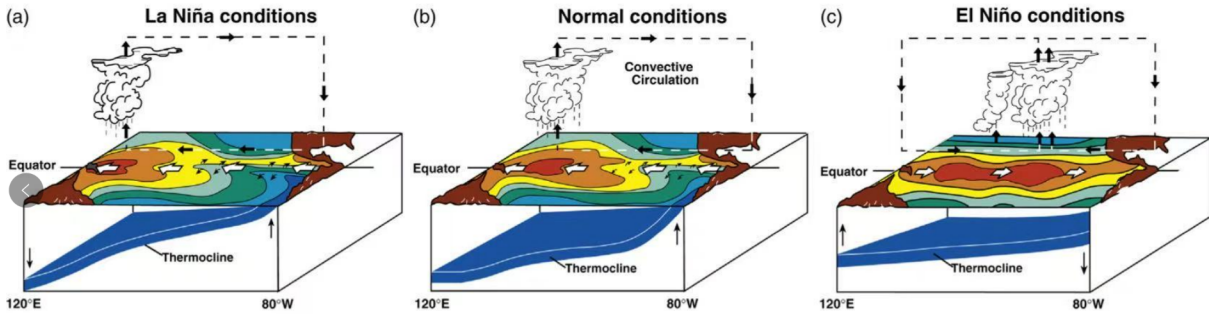


Figure 1: Schematic diagram showing the atmospheric and oceanic conditions along the equator in (a) La Niña, (b) Normal, and (c) El Niño conditions. ^[1]

A particularly fruitful framework is the low-order recharging oscillator originally proposed by Jin ^[2]. In his model, the SST deviation T and thermocline depth deviation h form a coupled pair

whose interaction drives ENSO cycles. In our review, we explore an extended model created by LIU Yudan^[3], by explicitly including a time-delay feedback term $\Delta = \kappa T(t - \tau)$ to represent the finite transit time of oceanic Rossby and Kelvin waves. These two waves are essential to the heat transfer in the Pacific Ocean, leading to the variation of the thermocline depth and the oscillations between El Niño and La Niña events.

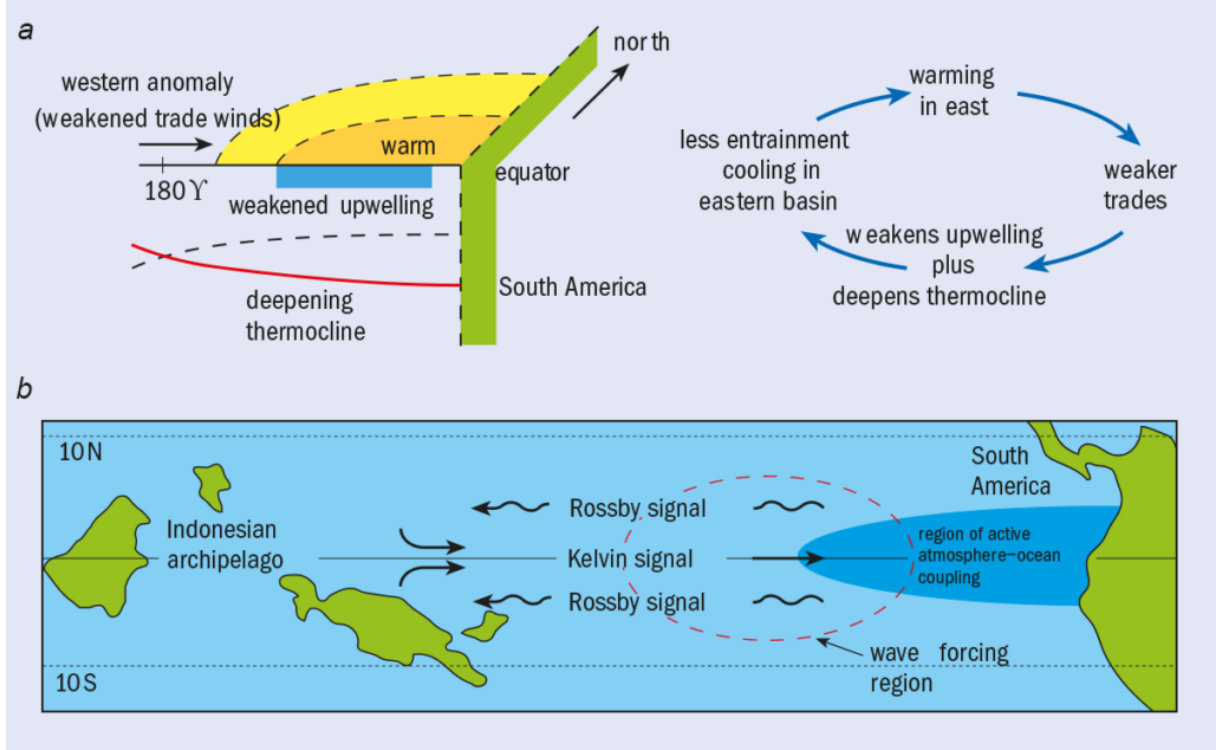


Figure 2: El Niño event formation and oceanic Rossby-Kelvin propagation – excerpted from the Physics World website (Fig. 4).^[4]

By carrying out a detailed analytical study—nondimensionalization, averaging, and linear stability analysis—we show that the inclusion of delay does not suppress ENSO; instead, it preserves a Hopf bifurcation to a stable limit cycle and even enriches the system’s dynamical regimes. In particular, varying the delay intensity κ and lag τ shifts the bifurcation threshold and modulates the amplitude and period of the resulting oscillations. We discuss these effects in the context of both El Niño and La Niña states, identifying parameter windows for super-critical and sub-critical bifurcations.

2 Materials and Methods

2.1 ENSO Recharge Oscillator: ENSO Recharge Oscillator with Time-Delayed Feedback (Δ)

We consider a recharge oscillator model^[2] for ENSO with time delay ^[3]

$$\begin{cases} \frac{dT}{dt} = CT + Dh - \varepsilon T^3 + \Delta, \\ \frac{dh}{dt} = -ET - R_h h. \end{cases} \quad (1)$$

- T : the East Pacific equatorial SST deviation.
- h : the thermocline depth deviation.
- C : the negative feedback coefficients in the oceanic thermocline.
- D : the Thermocline-SST coupling.
- E : the Wind-Thermocline feedback.
- R_h : the Bjerknes positive feedback coefficient.
- ε : the nonlinear damping coefficient.
- Δ : the delayed recharge term.

$$- \Delta = \kappa T(t - \tau)$$

– κ : the intensity of the delayed feedback.

– τ : the delay time.

$$\frac{dT}{dt} = CT + Dh - \varepsilon T^3 + \Delta, \quad (2)$$

$$\frac{dh}{dt} = -ET - R_h h, \quad (3)$$

Transforming the original two-dimensional system of equations into a univariate (only depends on T) second-order time-delay equation (Van der Pol-Duffing Oscillator Equation)

Taking time derivative for (2) we can get,

$$\frac{d^2T}{dt^2} = C \frac{dT}{dt} + D \frac{dh}{dt} - 3\varepsilon T^2 \frac{dT}{dt} + \frac{d\Delta}{dt}. \quad (4)$$

Substitute (3) into (4),

$$\begin{aligned} \frac{d^2T}{dt^2} &= C \frac{dT}{dt} + D(-ET - R_h h) - 3\varepsilon T^2 \frac{dT}{dt} - \frac{d\Delta}{dt} \\ \frac{d^2T}{dt^2} &= C \frac{dT}{dt} - DET - DR_h h - 3\varepsilon T^2 \frac{dT}{dt} + \frac{d\Delta}{dt} \end{aligned}$$

Since from (2),

$$\begin{aligned} \frac{dT}{dt} &= CT + Dh - \varepsilon T^3 + \Delta \\ -Dh &= CT - \varepsilon T^3 + \Delta - \frac{dT}{dt} \end{aligned}$$

Substitute $-Dh = CT - \varepsilon T^3 + \Delta - \frac{dT}{dt}$,

$$\begin{aligned} \frac{d^2T}{dt^2} &= C \frac{dT}{dt} - DET + \left(CT - \varepsilon T^3 + \Delta - \frac{dT}{dt}\right)R_h - 3\varepsilon T^2 \frac{dT}{dt} + \frac{d\Delta}{dt} \\ \frac{d^2T}{dt^2} &= \frac{dT}{dt} \left(C - R_h - 3\varepsilon T^2\right) + T(R_h C - DE) - \varepsilon R_h T^3 + \Delta R_h + \frac{d\Delta}{dt} \end{aligned}$$

Where

$$\begin{aligned} \mu\varepsilon &= R_h - C, \quad (\text{damping coefficient}) \\ \omega^2 &= DE - CR_h, \quad (\text{natural frequency without perturbation}) \end{aligned}$$

which implies

$$\begin{cases} -\mu\varepsilon = C - R_h, \\ -\omega^2 = CR_h - DE, \end{cases} \quad 0 < \varepsilon < 1, \quad CR_h < DE.$$

Recall that $\Delta = \kappa T(t - \tau)$ represents the cumulative effect of Rossby waves traveling westward, reflecting as Kelvin waves, and returning to the Central Pacific Ocean. The ENSO cycle (2-7 years) is much longer than the time delay τ (several months). Therefore, the instantaneous rate of change of the time delay feedback has a negligible contribution to the long-term evolution of the system.

Only its phase delay effect (i.e., Δ itself) needs to be considered; the term $\frac{d\Delta}{dt}$ could be ignored.

$$\frac{d^2T}{dt^2} = \frac{dT}{dt}(-\mu\varepsilon - 3\varepsilon T^2) - \omega^2 T - \varepsilon R_h T^3 + \Delta R_h, \quad \text{with } \mu\varepsilon = R_h - C, \quad \omega^2 = D E - C R_h$$

$$\frac{d^2T}{dt^2} = -\mu\varepsilon \frac{dT}{dt} - 3\varepsilon T^2 \frac{dT}{dt} - \omega^2 T - \varepsilon R_h T^3 + \Delta R_h$$

$$\frac{d^2T}{dt^2} + \varepsilon \mu \frac{dT}{dt} + 3\varepsilon T^2 \frac{dT}{dt} + \omega^2 T + \varepsilon R_h T^3 = \Delta R_h$$

To keep every term in the equation on a consistent scale and ensure that the delayed-feedback forcing enters only as a weak $O(\varepsilon)$ perturbation—rather than contaminating the leading $O(1)$ linear oscillation—we absorb the original Δ into ε . Concretely, we write

$$\Delta R_h = \varepsilon \Delta, \quad R_h = O(1), \quad \Delta = O(1), \quad \varepsilon \ll 1.$$

This rescaling aligns with the core assumptions of weakly nonlinear oscillation theory. It reflects that, physically, the delayed feedback acts as a secondary mechanism whose influence appears only at the perturbative level.

$$\frac{d^2T}{dt^2} + \omega^2 T + \varepsilon \left[(\mu + 3T^2) \frac{dT}{dt} + R_h T^3 \right] = \varepsilon \Delta$$

2.2 Van der Pol-Duffing Equation

Through mathematical transformations, we obtain the Van der Pol-Duffing equation for ENSO:

$$\frac{d^2T}{dt^2} + \omega^2 T + \varepsilon \left[(\mu + 3T^2) \frac{dT}{dt} + R_h T^3 \right] = \varepsilon \Delta \tag{5}$$

However, we notice that the Van der Pol-Duffing Equation has the form

$$\frac{dx^2}{dt^2} - \mu(1 - x^2) \frac{dx}{dt} + x = 0 \tag{6}$$

where μ is the damping term.

Van der Pol-Duffing Equation is a non-conservative, oscillating system with non-linear damping. If μ in (6) is less than 0 ($\mu < 0$), the system does not experience stable self-excited oscillations, but

asymmetric dynamical behavior between small amplitudes being dissipated and large amplitudes being amplified indefinitely. By comparing (6) with (5), we found that the sign before the damping term is opposite to the general form (6). This means when $\mu > 0$ in (5), the system will not perform stable oscillation if the time delay term Δ does not exist, vice versa.

To simplify the calculation, we set $\omega = 1$, and we know $\Delta = \kappa T(t - \tau)$

$$\frac{d^2T}{dt^2} + T = \varepsilon G\left(T, \frac{dT}{dt}, \Delta\right) \quad (7)$$

where

$$G\left(T, \frac{dT}{dt}, \Delta\right) = -\left(\mu + 3T^2\right)\frac{dT}{dt} - R_h T^3 + \kappa T(t - \tau).$$

To find the main oscillation, we temporally set $\varepsilon = 0$ (No damping) $\implies \frac{d^2T}{dt^2} + T = 0$

2.3 Polarization

In order to explain the oscillation more intuitively, we use the polarization method to solve the ODE above. Here, the amplitude A is the intensity of the El Niño/La Niña event (SST deviation); the phase ϕ reflects the timing characteristics of oscillations (El Niño to La Niña transitions).

$$\begin{aligned} \frac{d^2T}{dt^2} + T(t) &= 0 \\ r^2 + 1 &= 0 \\ r &= \pm\sqrt{-1} \\ r &= \pm i \\ T(t) &= A_1 \cos(t) + A_2 \sin(t) \end{aligned} \quad (8)$$

Polar form

$$\begin{cases} A_1 = A \cos(\phi) \\ A_2 = A \sin(\phi) \end{cases} \quad A = \sqrt{A_1^2 + A_2^2}, \phi = \arctan\left(\frac{A_2}{A_1}\right) \quad (9)$$

$$T(t) = A \cos(\phi) \cos(t) + A \sin(\phi) \sin(t)$$

$$T(t) = A \cos(t - \phi)$$

$$\begin{cases} x_1(t) = T(t), \\ x_2(t) = \frac{dT(t)}{dt}, \end{cases} \quad \text{Combining (7)} \Rightarrow \begin{cases} \frac{dx_1}{dt} = x_2(t), \\ \frac{dx_2}{dt} = -x_1(t) + \varepsilon G(x_1(t), x_2(t), T(t-\tau)). \end{cases} \quad (10)$$

$$\text{Hence: } x_1(t) = T(t) = A \cos(t - \phi), \quad x_2(t) = \frac{dT(t)}{dt} = -A \sin(t - \phi). \quad (11)$$

$$\begin{cases} T(t) = A \cos(t - \phi), \\ \frac{dT(t)}{dt} = -A \sin(t - \phi), \end{cases} \quad A = \sqrt{x_1^2 + x_2^2},$$

Now we have the solution form $T(t)$ to the ODE that could describe the periodical SST deviation oscillations. However, $A(t)$ and $\phi(t)$ are the slow varying function (ENSO 2 to 7 years), but the $\cos(t - \phi)$ term is the fast varying function (high frequency signals like season changing a full cycle over a single year). In order to separate the slow-fast varying functions, we here take the time derivative of the solution to get an expression that includes $\frac{dA}{dt}$ and $\frac{d\phi}{dt}$. Note that $\frac{dA}{dt}$ and $\frac{d\phi}{dt}$ are the ENSO main oscillation features respect to time.

To find $\frac{dA}{dt}$,

$$\begin{aligned} \frac{dA}{dt} &= \frac{1}{2} (x_1^2 + x_2^2)^{-\frac{1}{2}} \cdot \left(2x_1 \frac{dx_1}{dt} + 2x_2 \frac{dx_2}{dt} \right) \\ \frac{dA}{dt} &= \frac{2x_1 \frac{dx_1}{dt} + 2x_2 \frac{dx_2}{dt}}{2(x_1^2 + x_2^2)^{\frac{1}{2}}} \\ \frac{dA}{dt} &= \frac{x_1 \frac{dx_1}{dt} + x_2 \frac{dx_2}{dt}}{\sqrt{x_1^2 + x_2^2}} \end{aligned} \quad (12)$$

Since $\sqrt{x_1^2 + x_2^2} = A$

$$\frac{dA}{dt} = \frac{x_1 \frac{dx_1}{dt} + x_2 \frac{dx_2}{dt}}{A}$$

According to (10)

$$\begin{aligned}
\frac{dx_1}{dt} &= x_2, & \frac{dx_2}{dt} &= -x_1 + \varepsilon G, \\
\frac{dA}{dt} &= \frac{x_1 x_2 + x_2(-x_1 + \varepsilon G)}{A} \\
\frac{dA}{dt} &= \frac{\cancel{x_1 x_2} - \cancel{x_2 x_1} + x_2 \varepsilon G}{A} \\
\frac{dA}{dt} &= \frac{\varepsilon(-A \sin(t - \phi)) G}{A} \\
\frac{dA}{dt} &= -\varepsilon \sin(t - \phi) G.
\end{aligned} \tag{13}$$

To find $\frac{d\phi}{dt}$,

$$\begin{aligned}
\text{From (11)} \quad x_2 &= -A \sin(t - \phi), \\
\frac{dx_2}{dt} &= -\frac{d\phi}{dt} \sin(t - \phi) - A \cos(t - \phi) \left(1 - \frac{d\phi}{dt}\right),
\end{aligned}$$

and we know from (10)

$$\begin{aligned}
\frac{dx_2}{dt} &= -x_1 + \varepsilon G \\
&= -T + \varepsilon G \\
&= -A \cos(t - \phi) + \varepsilon G,
\end{aligned}$$

Equating the two expressions for $\frac{dx_2}{dt}$:

$$\begin{aligned}
-\frac{dA}{dt} \sin(t - \phi) - A \cos(t - \phi) \left(1 - \frac{d\phi}{dt}\right) &= -A \cos(t - \phi) + \varepsilon G, \\
-\frac{dA}{dt} \sin(t - \phi) - A \cos(t - \phi) + A \frac{d\phi}{dt} \cos(t - \phi) &= -A \cos(t - \phi) + \varepsilon G,
\end{aligned}$$

Cancelling the $-A \cos(t - \phi)$ terms:

$$A \frac{d\phi}{dt} \cos(t - \phi) - \frac{dA}{dt} \sin(t - \phi) = \varepsilon G$$

Here,

$$\frac{dA}{dt} = -\varepsilon G \sin(t - \phi)$$

Continuing,

$$\begin{aligned}
A \frac{d\phi}{dt} \cos(t - \phi) + \varepsilon G \sin^2(t - \phi) &= \varepsilon G, \\
A \frac{d\phi}{dt} \cos(t - \phi) &= \varepsilon G - \varepsilon G \sin^2(t - \phi), \\
A \frac{d\phi}{dt} \cos(t - \phi) &= \varepsilon G (1 - \sin^2(t - \phi)), \\
A \frac{d\phi}{dt} \cos(t - \phi) &= \varepsilon G \cos^2(t - \phi), \\
A \frac{d\phi}{dt} &= \varepsilon G \cos(t - \phi).
\end{aligned}$$

Now we have

$$\begin{cases} \frac{dA}{dt} &= -\varepsilon \sin(t - \phi) G \\ A \frac{d\phi}{dt} &= \varepsilon \cos(t - \phi) G \end{cases}$$

2.4 Averaging Method

Since we only want to focus on the main oscillation of the ENSO, we have to eliminate the high-frequency terms $\sin(t - \phi)$ and $\cos(t - \phi)$. To do that, we use the Averaging Method and substitute $T(t)$ and $\frac{dT}{dt}$ back to (7). The Averaging Method involves integrating $\frac{dA}{dt}$ and $\frac{d\phi}{dt}$ over $[0, 2\pi]$ to get an autonomous system that only related to $\frac{dA}{dt}$ and $\frac{d\phi}{dt}$. In this case, the main oscillation of ENSO does not related to the high-frequency terms, which could describe the slow varying feature of ENSO.

For function $\frac{dA}{dt}$, we have

$$\int_0^{2\pi} \frac{dA}{dt} dt = \frac{dA}{dt} \int_0^{2\pi} dt = 2\pi \frac{dA}{dt}.$$

Since $\frac{dA}{dt}$ is a slow varying function. It remains almost unchanged within an 2π oscillation cycle, so it can be treated as a constant over the integration. Then we have

$$\begin{aligned}
\int_0^{2\pi} \varepsilon \sin(t - \phi) G dt &= -\varepsilon \int_0^{2\pi} \sin(t - \phi) G dt. \\
G &= (-\mu + 3T^2) \frac{dT}{dt} + R_h T^3 + \kappa T(t - \tau).
\end{aligned}$$

Substitute $T = A \cos(t - \phi)$ and $\frac{dT}{dt} = -A \sin(t - \phi)$:

$$\begin{aligned}
G &= (-\mu + 3A^2 \cos^2(t - \phi))(-A \sin(t - \phi)) - R_h A^3 \cos^3(t - \phi) + \kappa A \cos^3(t - \phi - \tau), \\
&= (\mu - 3A^2 \cos^2(t - \phi))A \sin(t - \phi) - R_h A^3 \cos^3(t - \phi) + \kappa A \cos^3(t - \phi - \tau).
\end{aligned}$$

Let $\theta = t - \phi$:

$$G = (-\mu + 3A^2 \cos^2 \theta) A \sin \theta - R_h A^3 \cos^3 \theta + \kappa A \cos^3(\theta - \tau).$$

$$-\varepsilon \int_0^{2\pi} \sin(t - \phi) G dt = -\varepsilon \int_0^{2\pi} \sin \theta G d\theta.$$

$$\begin{aligned}
I_1 &= \int_0^{2\pi} \sin \theta (\mu + 3A^2 \cos^2 \theta) A \sin \theta d\theta, \\
&= \int_0^{2\pi} (\mu A \sin^2 \theta + 3A^3 \sin^2 \theta \cos^2 \theta) d\theta, \\
&= \mu A \int_0^{2\pi} \sin^2 \theta d\theta + 3A^3 \int_0^{2\pi} \sin^2 \theta \cos^2 \theta d\theta, \\
&= \mu A \int_0^{2\pi} \frac{1}{2} (1 - \cos 2\theta) d\theta + 3A^3 \int_0^{2\pi} \frac{1}{4} \sin^2(2\theta) d\theta, \\
&= \mu A (\pi) + 3A^3 \left(\frac{1}{4}\pi\right), \\
&= \pi A \left(\mu + \frac{3}{4}A^2\right).
\end{aligned}$$

$$\begin{aligned}
I_2 &= \int_0^{2\pi} \sin \theta (-R_h A^3 \cos^3 \theta) d\theta, \\
&= -R_h A^3 \int_0^{2\pi} \sin \theta \cos^3 \theta d\theta,
\end{aligned}$$

$$= -R_h A^3 \int_0^{2\pi} u^3 du, \quad \text{where} \quad \begin{cases} u = \cos \theta, \\ \frac{du}{d\theta} = -\sin \theta, \\ d\theta = -\frac{1}{\sin \theta} du. \end{cases}$$

$$\begin{aligned}
&= -R_h A^3 \left[\frac{u^4}{4} \right]_0^{2\pi} \\
&= -R_h A^3 \left[\frac{\cos^4 \theta}{4} \right]_0^{2\pi} \\
&= 0.
\end{aligned}$$

$$\begin{aligned}
I_3 &= \int_0^{2\pi} \sin \theta (kA \cos(\theta - \tau)) d\theta = kA \int_0^{2\pi} \sin \theta \cos(\theta - \tau) d\theta, \\
&= kA \int_0^{2\pi} \frac{1}{2} (\sin(2\theta - \tau) + \sin \tau) d\theta, \\
&= \frac{kA}{2} \int_0^{2\pi} \sin(2\theta - \tau) d\theta + \int_0^{2\pi} \sin \tau d\theta, \\
&= \frac{kA}{2} \left(\int_0^{2\pi} \frac{1}{2} \sin(u) du + \sin \tau [\theta]_0^{2\pi} \right), \quad \text{where } \begin{cases} u = 2\theta - \tau, \\ \frac{du}{d\theta} = 2, \\ d\theta = \frac{1}{2} du. \end{cases} \\
&= \frac{kA}{2} \left(\left[-\frac{\cos(2\theta - \tau)}{2} \right]_0^{2\pi} + 2\pi \sin \tau \right), \\
&= kA\pi \sin \tau.
\end{aligned}$$

$$\begin{aligned}
2\pi \frac{dA}{dt} &= -\varepsilon (I_1 + I_2 + I_3), \\
2\pi \frac{dA}{dt} &= -\varepsilon \left(\pi A \left(\mu + \frac{3}{4} A^2 \right) + 0 + \pi kA \sin \tau \right), \\
\frac{dA}{dt} &= -\frac{\varepsilon}{2} A \left(\mu + \frac{3}{4} A^2 + \kappa \sin \tau \right).
\end{aligned}$$

For function $\frac{d\phi}{dt}$, we have

$$\begin{aligned}
\int_0^{2\pi} \frac{d\phi}{dt} dt &= \frac{1}{2\pi} \frac{d\phi}{dt}. \\
\frac{1}{2\pi} \frac{d\phi}{dt} &= \int_0^{2\pi} \varepsilon \cos \theta G d\theta, \\
\frac{d\phi}{dt} &= \frac{1}{2\pi} \int_0^{2\pi} \varepsilon \cos \theta G d\theta.
\end{aligned}$$

$$\begin{aligned}
\int_0^{2\pi} \cos \theta (\mu + 3A^2 \cos^2 \theta) A \sin \theta d\theta &= 0, \\
\int_0^{2\pi} \cos \theta (-R_h A^3 \cos^3 \theta) d\theta &= -R_h A^3 \frac{3\pi}{4}, \\
\int_0^{2\pi} \cos \theta kA \cos(\theta - \tau) d\theta &= \pi kA \cos \tau.
\end{aligned}$$

$$\begin{aligned}\frac{A}{2} \frac{dA}{dt} &= \varepsilon \left(0 - R_h A^3 \frac{3\pi}{4} + \pi \kappa A \cos \tau \right), \\ \frac{d\phi}{dt} &= \pi A \varepsilon \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right), \\ \frac{d\phi}{dt} &= \frac{\varepsilon}{2} \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right).\end{aligned}$$

Hence, we have

$$\begin{cases} \frac{dA}{dt} = -\frac{\varepsilon}{2} A \left(\mu + \frac{3}{4} A^2 + \kappa \sin \tau \right) \\ \frac{d\phi}{dt} = \frac{\varepsilon}{2} \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right) \end{cases}$$

We know from (8)

$$T(t) = A \cos(t - \phi) = A_1 \cos(t) + A_2 \sin(t).$$

And recall from (9), we have

$$\begin{cases} A_1 = A \cos(\phi) \\ A_2 = A \sin(\phi) \end{cases} \quad A = \sqrt{A_1^2 + A_2^2}, \phi = \arctan\left(\frac{A_2}{A_1}\right)$$

Differentiating A_1 gives

$$\begin{aligned}\frac{dA_1}{dt} &= \frac{dA}{dt} \cos \phi - A \sin \phi \frac{d\phi}{dt} \\ &= -\frac{\varepsilon}{2} A \left(\mu + \frac{3}{4} A^2 + \kappa \sin \tau \right) \cos \phi - A \sin \phi \frac{\varepsilon}{2} \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right) \\ &= \frac{\varepsilon}{2} \left[\left(-\mu - \frac{3}{4} A^2 - \kappa \sin \tau \right) A \cos \phi - A \sin \phi \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right) \right] \\ &\text{with } A_1 = A \cos \phi, \quad A_2 = A \sin \phi, \quad A = \sqrt{A_1^2 + A_2^2}, \\ &= \frac{\varepsilon}{2} \left[A_1 \left(-\mu - \kappa \sin \tau \right) - \frac{3}{4} A^2 A_1 - \kappa A_2 \cos \tau + \frac{3}{4} R_h A^2 A_2 \right] \\ &= \frac{\varepsilon}{2} \left[A_1 \left(-\mu - \kappa \sin \tau \right) - \frac{3}{4} (A_1^2 + A_2^2) A_1 - \kappa A_2 \cos \tau + \frac{3}{4} R_h (A_1^2 + A_2^2) A_2 \right].\end{aligned}$$

Differentiating A_2 gives

$$\begin{aligned}
\frac{dA_2}{dt} &= \frac{dA}{dt} \sin \phi + A \cos \phi \frac{d\phi}{dt} \\
&= -\frac{\varepsilon}{2} A \left(\mu + \frac{3}{4} A^2 + \kappa \sin \tau \right) \sin \phi + A \cos \phi \frac{\varepsilon}{2} \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right) \\
&= \frac{\varepsilon}{2} \left[\left(-\mu - \frac{3}{4} A^2 - \kappa \sin \tau \right) A \sin \phi + A \cos \phi \left(\kappa \cos \tau - \frac{3}{4} R_h A^2 \right) \right] \\
&= \frac{\varepsilon}{2} \left[A_2 (-\mu - \kappa \sin \tau) - \frac{3}{4} A^2 A_2 + \kappa A_1 \cos \tau - \frac{3}{4} R_h A^2 A_1 \right] \\
&= \frac{\varepsilon}{2} \left[A_2 (-\mu - \kappa \sin \tau) - \frac{3}{4} (A_1^2 + A_2^2) A_2 + \kappa A_1 \cos \tau - \frac{3}{4} R_h (A_1^2 + A_2^2) A_1 \right].
\end{aligned}$$

Noting that the equilibrium point is $(A_1, A_2) = (0, 0)$ and dropping the nonlinear terms, the linearized system is

$$\begin{cases} \frac{dA_1}{dt} = \frac{\varepsilon}{2} \left(A_1 (-\mu - \kappa \sin \tau) - A_2 \kappa \cos \tau \right), \\ \frac{dA_2}{dt} = \frac{\varepsilon}{2} \left(A_2 (-\mu - \kappa \sin \tau) + A_1 \kappa \cos \tau \right). \end{cases} \quad (14)$$

2.5 Bifurcation analysis

The Jacobian matrix (ignoring the overall factor $\frac{\varepsilon}{2}$, which does not affect the characteristic equation) is

$$\mathbf{J} = \begin{pmatrix} -\mu - \kappa \sin \tau & -\kappa \cos \tau \\ \kappa \cos \tau & -\mu - \kappa \sin \tau \end{pmatrix}. \quad (15)$$

$$\det(\mathbf{J}(A_1, A_2) - \lambda \mathbf{I}) = 0$$

$$\det(\mathbf{J}(0, 0) - \lambda \mathbf{I}) = 0$$

$$\det \begin{pmatrix} -\mu - \kappa \sin \tau - \lambda & -\kappa \cos \tau \\ \kappa \cos \tau & -\mu - \kappa \sin \tau - \lambda \end{pmatrix} = (-\mu - \kappa \sin \tau - \lambda)^2 + (\kappa \cos \tau)^2 = 0. \quad (16)$$

$$(-M - \kappa \sin \tau)^2 - 2\lambda(-M - \kappa \sin \tau) + \lambda^2 + (\kappa \cos \tau)^2 = 0,$$

$$\lambda^2 + 2\lambda(M + \kappa \sin \tau) + (M + \kappa \sin \tau)^2 + (\kappa \cos \tau)^2 = 0.$$

$$\lambda = \frac{-2(\mu + \kappa \sin \tau) \pm \sqrt{(\mu + \kappa \sin \tau)^2 - 4[(\mu + \kappa \sin \tau)^2 + (\kappa \cos \tau)^2]}}{2},$$

$$\Delta = 4(\mu + \kappa \sin \tau)^2 - 4\left[(\mu + \kappa \sin \tau)^2 + (\kappa \cos \tau)^2\right] = \underbrace{4(\mu + \kappa \sin \tau)^2}_{>0} - \underbrace{4\left[(\mu + \kappa \sin \tau)^2 + (\kappa \cos \tau)^2\right]}_{>0}.$$

Note that

$$4(\mu + \kappa \sin \tau)^2 < 4\left[(\mu + \kappa \sin \tau)^2 + (\kappa \cos \tau)^2\right],$$

therefore

$$\Delta < 0 \quad \text{for all real parameters.}$$

Moreover, in our delay-differential ENSO model, the feedback time is set by the equatorial wave transit time

$$\tau = \frac{L}{c},$$

where $L \approx 1.5 \times 10^7$ m is the zonal width of the Pacific and $c \approx 0.8$ m/s is the Rossby–Kelvin wave speed. Numerically,

$$\tau \approx \frac{1.5 \times 10^7}{0.8} \text{ s} = 1.875 \times 10^7 \text{ s} \approx 217 \text{ days} \approx 7.2 \text{ months}.$$

By contrast, observed ENSO have a period of roughly 48 months. Because

$$\tau \approx 7.2 \text{ months} \implies \tau \bmod \pi \neq \frac{\pi}{2}.$$

We have $\cos \tau \neq 0$ and hence

$$(\kappa \cos \tau)^2 > 0$$

In the characteristic equation. This guarantees that the solutions for λ are complex (have nonzero imaginary parts), resulting in either a center or a spiral.

$$\text{Re}(\lambda) = -(\mu + \kappa \sin \tau),$$

where $\mu + \kappa \sin \tau$ plays the role of an effective damping parameter. Consequently:

- If $\mu + \kappa \sin \tau > 0$, then $\text{Re}(\lambda) < 0$ and the equilibrium is a *stable spiral*.
- If $\mu + \kappa \sin \tau = 0$, then $\text{Re}(\lambda) = 0$ and the equilibrium is a *center*.
- If $\mu + \kappa \sin \tau < 0$, then $\text{Re}(\lambda) > 0$ and the equilibrium is an *unstable spiral*.

In the absence of delay, ENSO require $\mu < 0$. However, the time delay introduced by the Rossby-to-Kelvin wave feedback can sustain oscillations even when $\mu > 0$.

3 Results

We here will verify the bifurcation results obtained above to show the ENSO Recharging Oscillator Model with time-delay with simple numerical simulations.

Consider $\varepsilon = 0.1$, $R_h = 1$, $\mu = 2 > 0$

- If we fixed the delay time $\tau = 2a$ (2 years)
 - for smaller $|\kappa|$ ($\kappa = -1$)

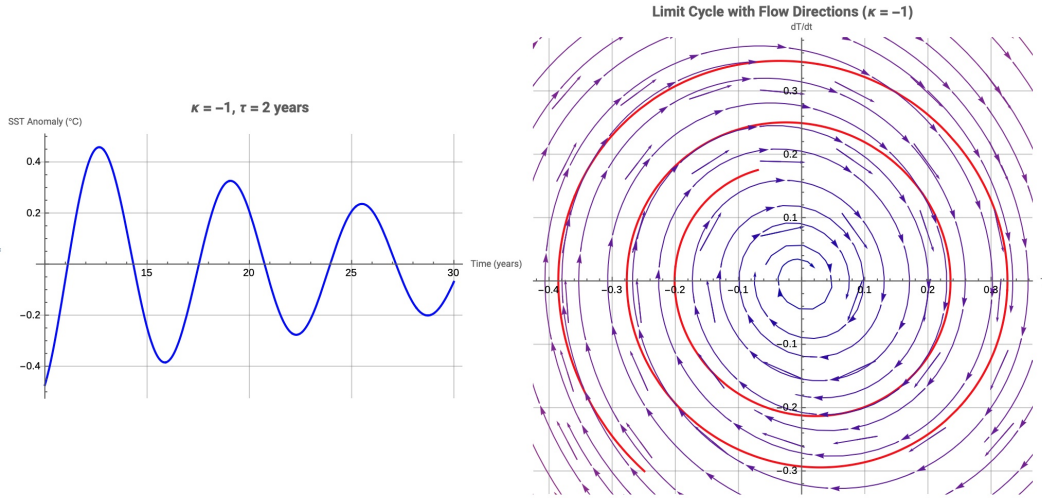


Figure 3: Time sequence and phase plot when $\kappa = -1$, $\tau = 2a$

We noticed that for the left graph (time sequence), the amplitude SST anomaly is decreasing over time. The phase plot indicates that the origin is a stable focus, and the trajectory converges inward along the spiral (red line), which represents a stable spiral. Physically, this case indicates insufficient negative feedback, resulting in self-excited energy dissipation through damping that decays the ENSO. The ENSO will eventually disappear.

– for greater $|\kappa|$ ($\kappa = -2.6$)

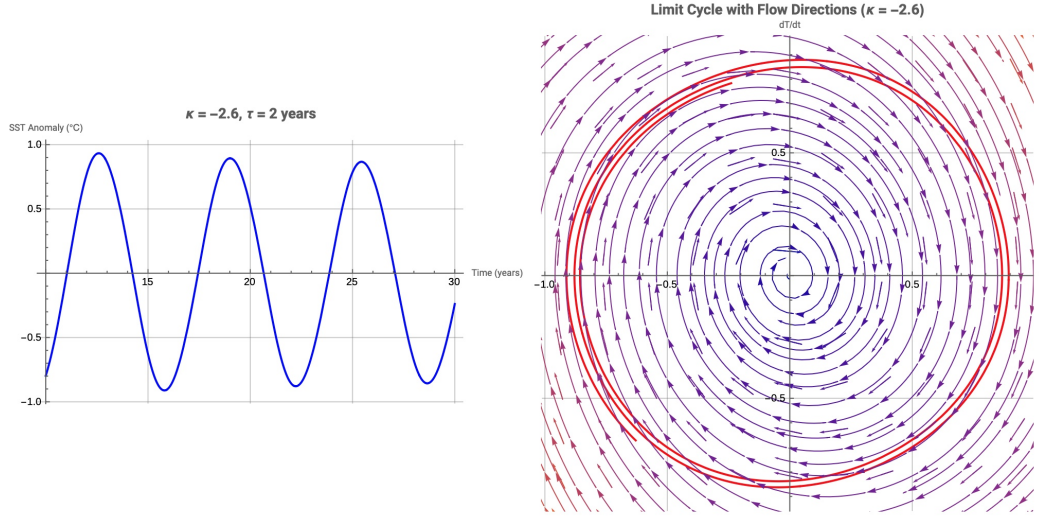


Figure 4: Time sequence and phase plot when $\kappa = -2.6$, $\tau = 2a$

We noticed that for the left graph (time sequence), the SST anomaly is forming almost a perfect sinusoidal pattern over time, and the amplitude almost does not change. The phase plot shows that the origin becomes an unstable focus, but the trajectory no longer converges to the equilibrium point, but spirals outward into a closed red limit cycle. Physically, this case means Negative feedback is enhanced, and self-excitation energy exceeds damping, which supports ENSO continuously over the year.

– for even greater $|\kappa|$ ($\kappa = -10$)

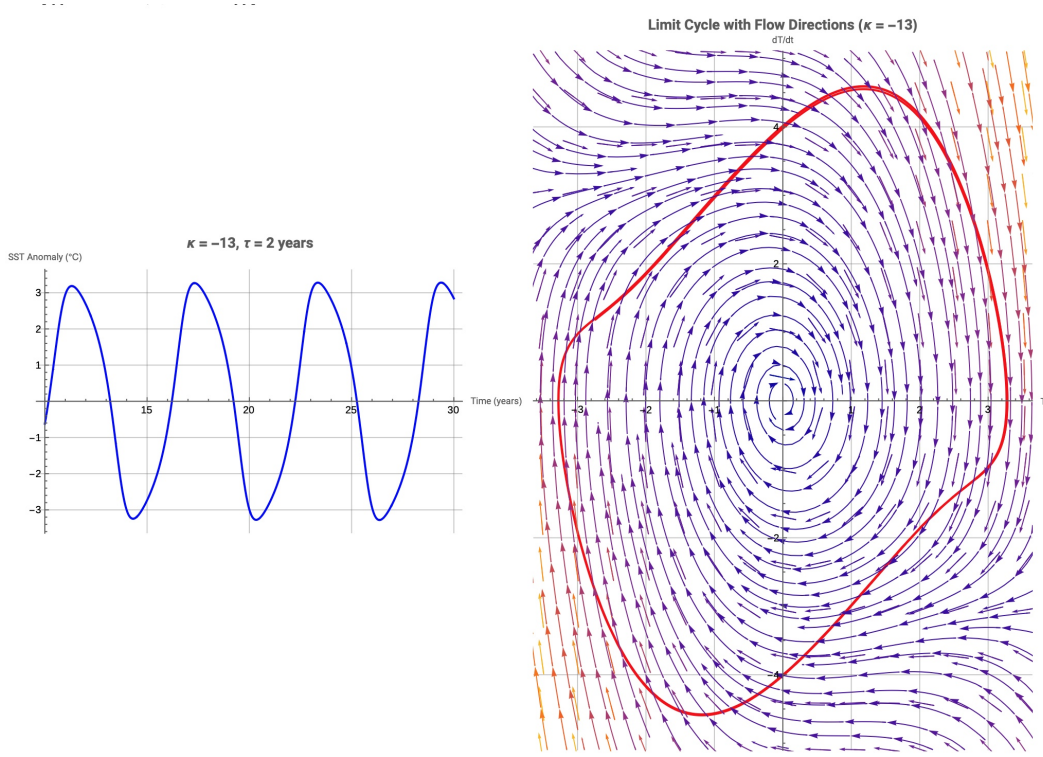


Figure 5: Time sequence and phase plot when $\kappa = -13$, $\tau = 2a$

Although we are expecting an unstable spiral for the phase plot, the nonlinear term $-\varepsilon T^3$ in (1) limits the infinite growth of the SST deviation and then forms a limited cycle. In this case, we could lower the initial SST deviation perturbation $T = 0.1$ and the delay time $\tau = 2$ to get an unstable graph. However, for Figure 5 above, we could notice that as the intensity of the delayed feedback increases, the change in SST deviation becomes more rapid, and stays at a higher amplitude (SST deviation) for a longer time.

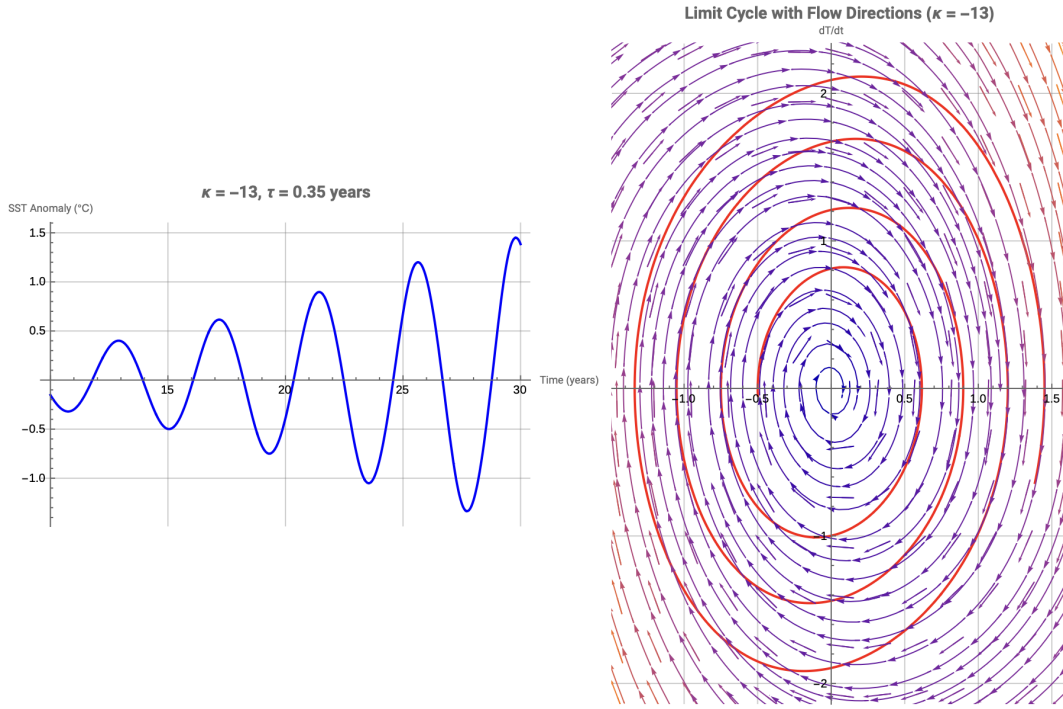


Figure 6: Time sequence and phase plot when $\kappa = -13, \tau = 0.35a$

Now, we can clearly see the amplitude of the SST deviation on the time-sequence graph getting greater as time progresses. The phase plot shows an unstable spiral until it reaches a limited cycle, as seen in Figure 5. Note that the infinite growth situation is not realistic, as the ocean-atmosphere system is not limited to the Pacific area but extends around the globe. The adjustment by oceanic/atmospheric circulation will prevent "super El Niño/La Niña events" from happening

- If we fixed the intensity of the delayed feedback $\kappa = -2.6$
 - $\tau = 2a$ (years)

We analyzed in Figure 4

- $\tau = 3a$ (years) Compared to Figure 4, we can see that κ also influences the mode of

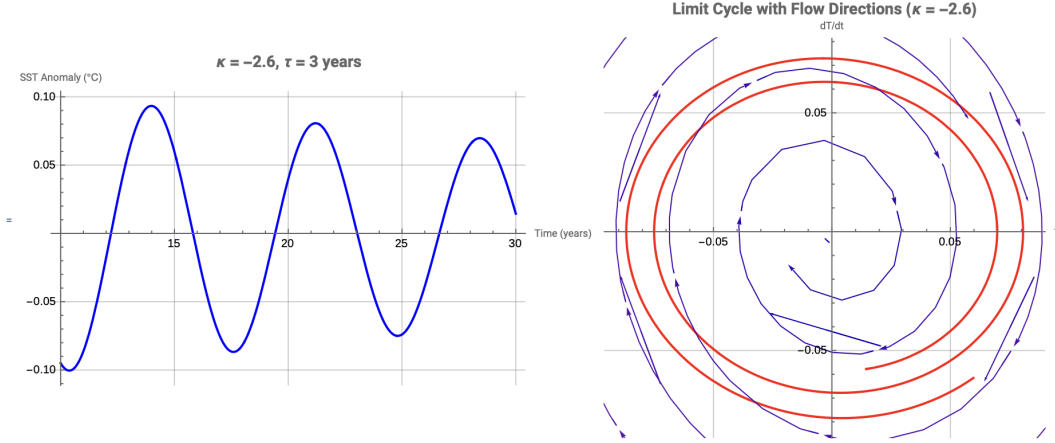


Figure 7: Time sequence and phase plot when $\kappa = -13$, $\tau = 3a$

ENSO. In this case (Figure 7), the situation is similar to the stable spiral case in Figure 3.

From the simple numerical simulation above, we have verified that the delay time τ and the intensity of delayed feedback κ indeed affect the ENSO modes even when the damping term $\mu > 0$.

4 Discussion

By introducing the time-delay term to the traditional ENSO model, we found that even when the oscillator damping is positive ($\mu > 0$), the ENSO still can exist, and highly sensitive on the intensity of the time-delay feedback κ and the delayed time τ .

To be specific, the Hopf bifurcation of the ENSO mode depends on the term $\mu + \kappa \sin \tau$ when it is less than, equal to, and greater than 0. For the unstable spiral case, $\mu + \kappa \sin \tau < 0$, we have observed that in actual data, no "super El Niño" events with infinite growth have been reported. Our model suggests that this may be due to the effect of the nonlinear damping term ($-\varepsilon T^3$), which limits the infinite growth of SST anomalies and causes the system to approach a finite-amplitude limit cycle

(Figure 5) over time. This mechanism is consistent with the observed upper limit of ENSO event amplitude, indicating that nonlinear effects play a crucial role in regulating ENSO intensity.

Nevertheless, our investigations did have some limitations. We ignored the influence of high-frequency seasonal forcing (such as annual cycles) in the Averaging Method, while in reality, ENSO is modulated by seasons. Our simulation didn't carefully select the values for each variable obtained from the real observation. In the context of climate change, the delayed time and intensity of ENSO may change, which in turn affects its oscillatory properties. For example, if global warming leads to faster Rossby wave propagation (increase in c), $\tau = \frac{L}{c}$ may decrease, thus changing the bifurcation behavior of ENSO. Furthermore, the mathematical modeling didn't consider the asymmetrical feature of the equatorial east-west Pacific Ocean, which ignored the geographical influence on actual ENSO.

Regarding the limitations discussed above, we could take another step further for our investigation in the following aspects. Multiscale interactions are essential in interannual ENSO, so coupling effects of delayed feedback with seasonal forcing and interannual variability could be considered. As the oscillation is dynamic and sensitive to several oceanic and atmospheric parameters, we could investigate how global warming affects ENSO dynamics by changing τ . In the numerical simulation, we can utilize historical ENSO event statistics κ and τ to enhance the predictive power of the model.

5 Conclusion

In conclusion, this study demonstrates through theoretical analysis and numerical simulations that time-delayed feedback of oceanic Rossby-Kelvin wave is one of the key mechanisms of self-sustained oscillations in ENSO, even when the damping term $\mu > 0$, providing a new perspective for understanding the nonlinear dynamics of ENSO.

This paper expanded the traditional ENSO recharging oscillation model proposed by Jin ^[2] by introducing the time-delay term $\Delta = \kappa T(t - \tau)$ for oceanic Rossby-Kelvin wave.

$$\begin{cases} \frac{dT}{dt} &= CT + Dh - \varepsilon T^3 + \Delta, \\ \frac{dh}{dt} &= -ET - R_h h. \end{cases}$$

We converted the 2-dimensional system of ODEs into the Van der Pol-Duffing Equation since the ENSO is a non-linear oscillation with damping.

$$\frac{d^2T}{dt^2} + \omega^2 T + \varepsilon \left[(\mu + 3T^2) \frac{dT}{dt} + R_h T^3 \right] = \varepsilon \Delta$$

We polarized the general form solution of the equation and separated the slow-varying functions for A and ϕ to find the main oscillation of ENSO. In this case, the fast-varying terms, such as seasonal forcing, have been ignored using the average method. We eventually found the Hopf bifurcation at the component amplitudes equilibrium point $(A_1, A_2) = (0, 0)$ by using the Jacobian Matrix and eigenvalues to formulate the mode of ENSO.

$$\begin{cases} \frac{dA_1}{dt} = \frac{\varepsilon}{2} \left(A_1(-\mu - \kappa \sin \tau) - A_2 \kappa \cos \tau \right), \\ \frac{dA_2}{dt} = \frac{\varepsilon}{2} \left(A_2(-\mu - \kappa \sin \tau) + A_1 \kappa \cos \tau \right). \end{cases}$$

The eigenvalues were complex and the bifurcation behavior was dependent on the expression $\mu + \kappa \sin \tau$ and highly sensitive to the intensity of the delayed feedback κ and delayed time τ even when the damping term $\mu > 0$

$$\mu + \kappa \sin \tau > 0 \leftarrow \textbf{stable spiral}$$

$$\mu + \kappa \sin \tau = 0 \leftarrow \textbf{center}$$

$$\mu + \kappa \sin \tau < 0 \leftarrow \textbf{unstable spiral}$$

A limited cycle will appear when $\mu + \kappa \sin \tau = 0$ and $\mu + \kappa \sin \tau > 0$. As for the unstable spiral, the non-linear term $(-\varepsilon T^3)$ will limit the infinite growth of SST deviation, which is consistent with the realistic observation.

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